



Variable DC Power Supply

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Abstract

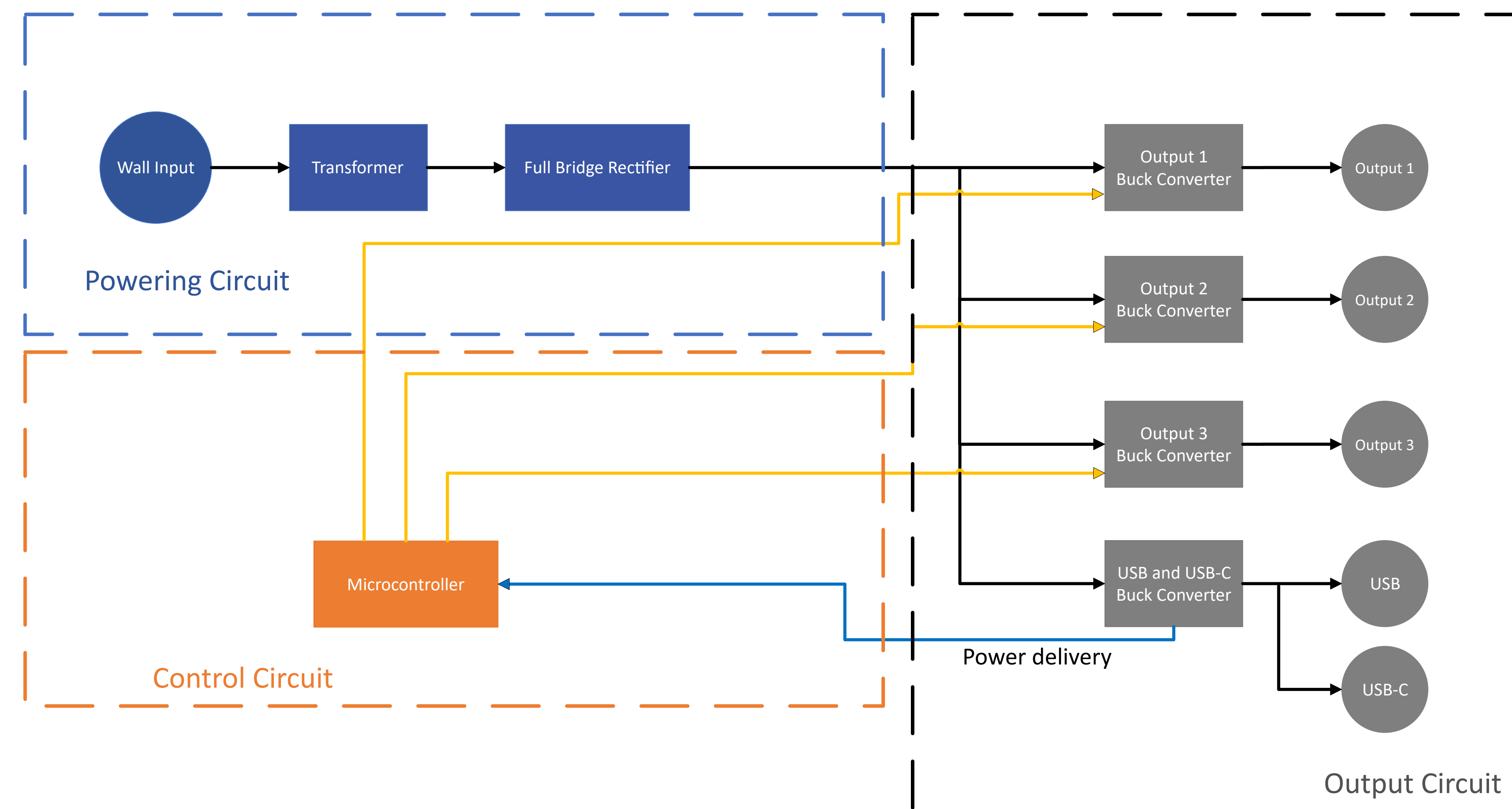
This project presents the design and development of a custom variable DC power supply intended for versatile laboratory and prototyping applications. The system has five different outputs: three variable channels that deliver up to 30V at 3A through banana connectors, one USB 2.0, and one USB-C output for powering small electronics. To ensure high efficiency, stable regulation, and compact implementation, each variable channel is driven by its own buck converter, while the USB and USB-C ports share a converter. The entire system is powered from a standard wall outlet, stepped down through a transformer, and then a rectifier to provide DC power for the buck converters. A Teensy 4.0 microcontroller is used for user interface management. User interaction is supported through an integrated display, keypad, and hardware on/off switches.

Specifications

The variable DC power supply was built to meet the following requirements/specifications:

- Five outputs: three variable, one USB 2.0, one USB-C
- Variable output voltage rating: 3-30V $\pm 5\%$
- Variable output current rating: 3A max
- Variable outputs are controlled by a user via keypad with voltage values displayed on a screen
- All outputs can be enabled/disabled
- Device is powered from a standard wall outlet
- Size: standard benchtop equipment

System Overview

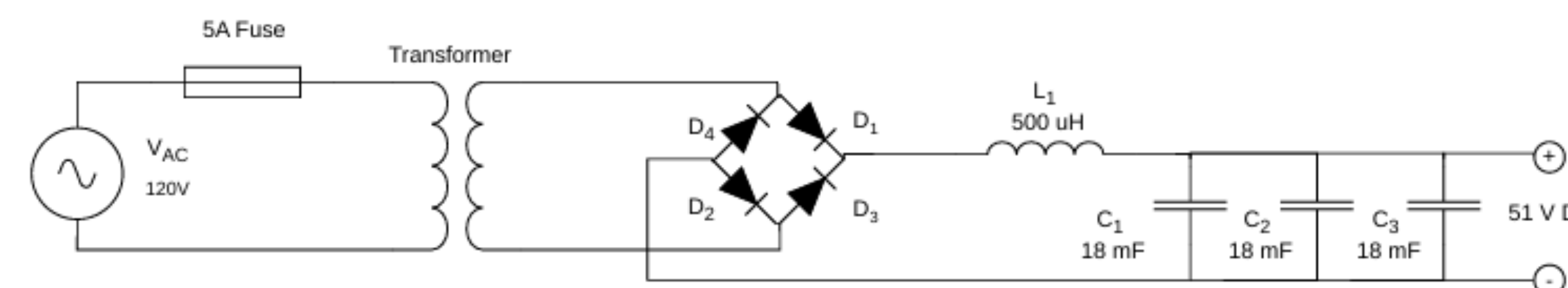


System Block Diagram

System is composed of three main parts:

Powering Circuit:

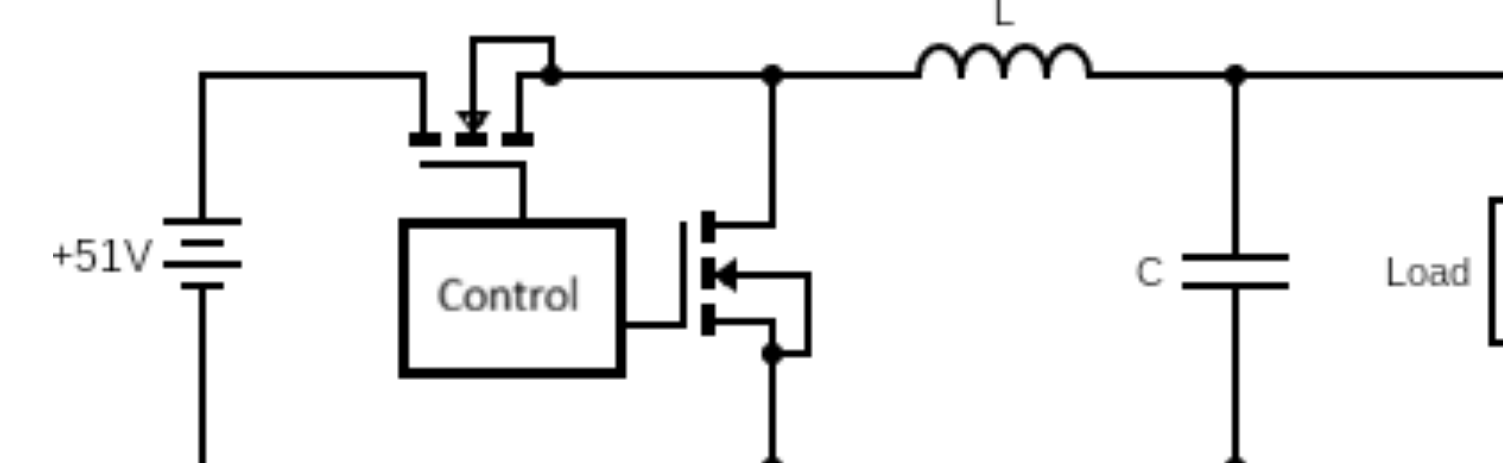
Supplies steady $\sim 51V$ DC to the buck converters. A 540W transformer steps down 120V AC from a wall plug down to 36V AC. A full bridge rectifier converts 36V AC into $\sim 51V$ DC to be used as the input voltage to the buck converters.



Powering Circuit Schematic

Output Circuit:

Steps down the 51V DC from the powering circuit to variable 3-30V DC for the output of the device. This circuit consists of four buck converters. Three of the buck converters are variable, rated for an output of 3 – 30V at 3A. The fourth buck converter is a fixed 5V rated up to 4A.



Output Buck Converter

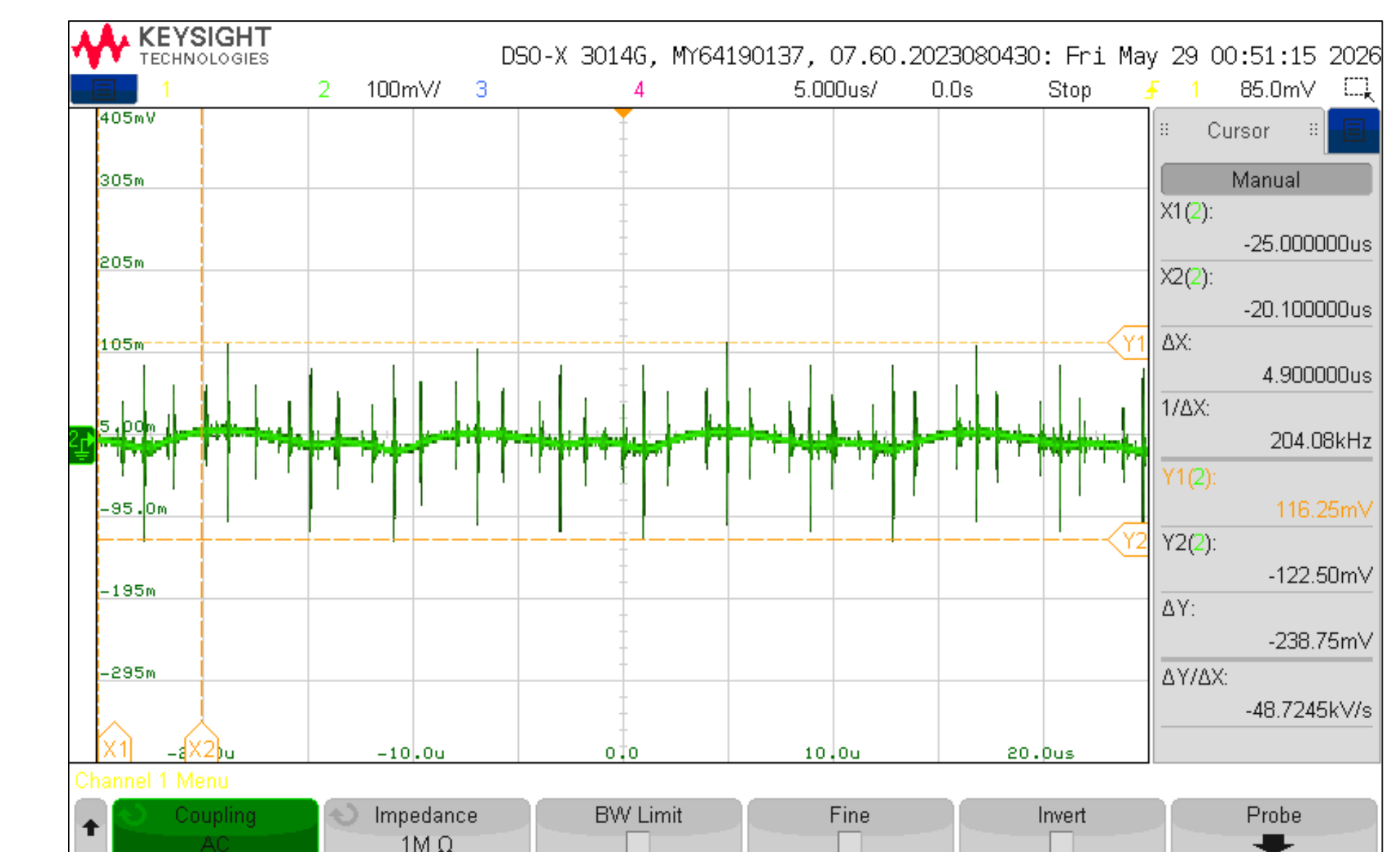
Control Circuit:

Controls the output of the buck converters and the user interface. Buck converter output is controlled via a controller IC and digital potentiometer. User interface is controlled via a screen and keypad.

Results

The variable DC power supply was plugged into the wall and tested using all three channels. The waveform below shows the output of channel two under these conditions:

- All channels set to the same values
- 3 Ω purely resistive load
- 9V/3A output



The table below shows the measured output results from all three channels on simultaneously:

Resistive Load					
Channel	Voltage	Load Current	Load Resistance (Ohm)	Voltage Ripple	% Ripple
1	9V	3A	3 Ω	198mV	2.20%
2	9V	3A	3 Ω	238mV	2.64%
3	9V	3A	3 Ω	385mV	4.30%

Overall, all outputs were within 5% voltage ripple except for low voltages of 3V-3.3V. The voltage ripple stayed around the same as voltage increased, therefore the voltage ripple at for example 30V was as low as 0.28%. To conclude, the project meets the required specifications except the low voltage ripple and conditions not yet able to be tested.

Considerations/Future Work

- Variable current limiting for output channels
- Adding integrated PCB On/Off switching
- Reducing transformer size and ratings
- Add SYNC pins for IC to control switching frequency
- Minimized losses and increase efficiency



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